

Introduction and Motivation

Dynamic point clouds are a core technology in the creation of volumetric video. However, due to the dense geometry and per-point attributes that evolve over time, raw dynamic point clouds are very bandwidth intensive and take up a large amount of storage. Therefore, compression is required in applications that involve dynamic point clouds in storage, transmission, and real-time streaming.

In recent years, multiple compression algorithms have been developed to address this issue. These include Video-based Point Cloud Compression (V-PCC), Geometry-based Point Cloud Compression (G-PCC), Google's Draco, Meta's openLZ, and more recently neural point-cloud compression models. Comparing these algorithms fairly on dynamic point-cloud sequences is challenging because their performance is strongly affected by geometric properties of the input data such as scale, sampling density, noise, and temporal consistency.

This project aims to implement a reproducible evaluation framework for dynamic point cloud compression, with an emphasis on geometry-aware preprocessing and analysis. The project will implement a set of geometric preprocessing operators from scratch that modify the point cloud before compression and allow a controlled study of how geometric regularization affects compression efficiency and reconstruction quality. The goal is to investigate how normalization, resampling, and smoothing of point cloud geometry influence rate-distortion performance and temporal stability across different codec families.

The project is guided by two main research questions. First, how do geometry-aware preprocessing steps such as normalization, density regularization, and MLS smoothing influence the rate-distortion performance of classical and neural point cloud compression algorithms on dynamic sequences? Second, to what extent do these preprocessing steps improve temporal stability and reconstruction consistency across frames?

Methodology and Technical Approach

Experiments will be conducted on publicly available dynamic point cloud datasets such as the 8i Voxelized Full Bodies dataset, which contains temporally evolving human sequences with color attributes.

The project will implement preprocessing modules from scratch and apply them to dynamic point cloud sequences prior to compression. The first component will be

geometric normalization, which will include centering the cloud around its centroid, scaling it to a consistent bounding box, and optionally applying PCA-based alignment to normalize orientation. This ensures that different datasets are represented in a comparable coordinate space and reduces sensitivity of compression algorithms to scale and global position.

The second component will address sampling density. Point clouds captured from sensors often contain uneven or redundant sampling, which can negatively affect compression efficiency. To address this, the project will implement density normalization using voxel-based resampling or radius-based neighborhood sampling. These methods will reduce redundancy in densely sampled regions while maintaining overall geometric structure, allowing the study of how spatial regularity influences compression results.

The third preprocessing operator will implement a simplified Moving Least Squares (MLS) smoothing approach. MLS smoothing estimates a locally fitted surface from neighboring points and projects points toward this surface, reducing noise while preserving the underlying shape. This step aims to investigate how geometric noise and small surface irregularities influence compression bitrate and reconstruction quality, especially in dynamic sequences where noise may vary across frames.

To ensure fair comparison across codecs, all point clouds will be normalized to a consistent coordinate scale prior to compression, and identical preprocessing parameters will be used across all methods for a given experiment. Rate control will be standardized by selecting comparable bitrate ranges or quality parameters for each codec so that results are evaluated at similar operating points. Resampling and smoothing parameters will also be fixed per experiment to avoid introducing codec-specific bias. Pretrained neural models will be selected from publicly available implementations, and their versions, training datasets (if documented), and configuration settings will be recorded so that results remain reproducible and comparable.

After preprocessing, the modified point clouds will be passed to multiple compression methods including V-PCC, G-PCC, Draco, openLZ, and selected pretrained neural models. The evaluation will measure bitrate, compression ratio, encoding and decoding time, and reconstruction quality using geometry-based metrics such as point-to-point (D1) and point-to-plane (D2) PSNR. Metrics will be computed per frame and aggregated across sequences in order to capture both spatial accuracy and temporal stability. By comparing results obtained from different preprocessing pipelines, the project will analyze how geometric normalization, density regularization, and smoothing affect the rate-distortion behavior of each codec.

In addition to per-frame distortion metrics, temporal stability will be evaluated by measuring the variation of reconstruction error between consecutive frames and by computing sequence-level statistics such as the standard deviation of frame-wise geometry PSNR. This allows detection of temporal instability or flickering effects that may not be visible in averaged metrics alone.

Together, these experiments will provide insight into how geometric preprocessing interacts with compression performance in dynamic point cloud pipelines.